



FACULTY OF COMPUTING

SEMESTER 1/20252026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 03

PROJECT 1:

AI PROPOSAL

THEME:

HEALTHCARE

LECTURER:

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Group no: Group 4

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Problem Statement: Triage Bottlenecks and Misdiagnosis in Emergency Rooms

High patient volumes and scarce medical resources are placing an increasing strain on emergency departments (EDs) worldwide, resulting in extended wait times and postponed treatment. Prioritizing patients according to the severity of their conditions is largely dependent on the triage process, which is frequently carried out by non-specialist staff. Triage assessment errors, however, may lead to incorrect diagnoses or underestimations of critical cases, delaying treatment and possibly worsening the condition of some patients with severe conditions. In addition to endangering patient safety and results, this inefficiency puts further burden on hospital resources and emergency medical teams.

AI Solution & Goal of AI Solution

The proposed AI solution, known as the AI-Powered Intelligent Triage Assistant (AITA) is an application prototype that is designed to revolutionize how emergency departments handle evaluates and categorize patients based on the urgency of their medical condition to prioritize their care known as triage assessment.

The AITA system leverages artificial intelligence and machine learning algorithms to analyze patients through symptoms, vital signs, and medical history in real time analysis. The systems is able to generate a severity score through all these informations and immediately recommends the appropriate next steps for treatment. By incorporating IoT-enabled medical devices such as digital thermometer, the system can collect input with enhance accuracy and precision.

This approach will reduce human intervention during analysis which could vary in time and accuracy depending on the staff experience and the severity during the triage assessment. Through predictive analytics and pattern recognition, the system assists triage nurses in identifying potential red flags for critical conditions that may otherwise be overlooked. Therefore, AITA serves as a decision-support tool that enhances efficiency, accuracy, and patient safety within emergency departments.

Ultimately, the goal of implementing AITA is to transform emergency care into a more efficient and accurate system. Specifically, the AI solution aims to:

- **Reduce critical patient waiting times** by quickly identifying and prioritizing severe cases for immediate attention.
- **Improve diagnostic accuracy and consistency** through AI-powered analysis that minimizes human bias and fatigue.
- **Enhance workflow efficiency** by automating repetitive triage tasks and allowing healthcare staff to focus on direct patient care.
- **Improve overall patient outcomes** through faster, more accurate, and timely treatment interventions.

AI Solution	AI-Powered Intelligent Triage Assistant (AITA): A mobile/web application prototype designed to assist triage nurses by analysing patient symptoms, vital signs, and medical history to provide an immediate, ranked severity score and recommend the next steps.
The Goal of AI Solution	To reduce critical patient waiting times and increase the accuracy and consistency of initial triage severity assignment in emergency departments, thereby improving patient outcomes.

Process of **Emphasis** in Design Thinking

In the empathize stage, we need to get some time and comprehend the needs, challenges, and feelings of our users. The absence of empathy means that our project will not have the essential user-centric approach that usually defines the success or failure of a product. Hence, it is necessary that we understand the actual lives of the emergency room employees and patients in order to guarantee that our artificial intelligence solution will actually cover their pain points and positively transform their daily processes. To accomplish this, we investigated several guiding questions to better understand our users' perspectives and emotions in actual emergency department settings.

WHO are we empathizing with?

Our target users are triage nurses and emergency room doctors who are responsible for assessing patient severity during emergencies. They must make fast, accurate judgments under pressure. Secondary stakeholders include patients, who often experience long waiting times, and hospital administrators, who aim to optimize resource allocation and ensure smooth workflow in the emergency department.

What do they need to DO?

Triage nurses must quickly evaluate patients based on symptoms, medical history, and vital signs. They must determine the severity level and prioritize treatment within minutes. To facilitate this process, they require an intelligent system capable of analysing patient data in real time, providing a reliable severity score, and recommending next steps to ensure consistent and accurate triage decisions.

What do they SEE?

They see overcrowded waiting spaces, distressed patients, and an ongoing number of new cases, particularly during peak hours. They also see several computer screens and paperwork, which can be overwhelming. Nurses frequently report that manual triage assessments are inconsistent due to time constraints and fatigue.

What do they SAY?

They often say things like:

- “There are too many patients and not enough time.”
- “I’m afraid of missing a critical case.”
- “We need a faster, more reliable way to identify emergencies.”

This indicates that they are concerned with patient safety, high workload, and better decision-support tools.

What do they DO?

They multitask in cases of evaluating new patients, following up on the old ones, updating the records, and communicating with physicians. They take quick decisions during busy shifts depending on experience and little information available, thus putting high risk of human error.

What do they HEAR?

They hear constant alerts, patient complaints, and instructions from doctors. They also receive administrative messages regarding efficiency, wait time goals, and lack of resources. This constant aural stimulation contributes to their psychological strain and work pressure.

What do they THINK AND FEEL?

They are stressed, anxious, pressured to make quick but correct decisions. Most of them are afraid of misdiagnosing or underestimating the condition of a patient, and this may very well cause serious consequences. In spite of such pressures, they also experience a high degree of responsibility and commitment to patient safety. They wish that they had a system that would help them to make judgments, minimize errors and relieve their cognitive load at the critical time.

Process of **Define** in Design Thinking

User	Need	Insight
A Triage Nurse who works in the emergency department and needs to quickly assess patients' severity levels under pressure.	• Need 1: To have an intelligent system that can instantly cross-reference uncommon symptom combinations with a large knowledge base and provide a quantifiable risk score (eg,1-5). • Need 2: To receive real-time alerts that flag critical warning signs from connected monitoring devices.	• First Insight: Due to patient volume and time limits, nurses are overwhelmed; an AI system that automates part of the triage process can improve both speed and accuracy. • Second Insight: Real-time data analysis ensures that crucial patients are appropriately prioritised and helps avoid human error. • Third Insight: A user-friendly interface with clear action recommendations can reduce stress and decision fatigue during peak hours.

TASK DISTRIBUTION

Task	Description	PIC	Deadline
Describe the process of Emphases in DT	Problem and pain, user/stakeholder and the goal are clearly described. Who is my user? What matters to them?	LAU YEE WEN	27/10/2025
Describe the process of Define in DT	Covers user/stakeholder needs in-depth with details and examples. What are their needs?	CHERYL CHEONG KAH VOON	27/10/2025
The goal of AI solution	AI solution shows large amount of original thought. Ideas are creative and inventive.	MUHAMMAD ADAM BIN RAZALI	27/10/2025
Name AI solution	AI-Powered Intelligent Triage Assistant (AITA)	ELIJAH SHE YU SHENG	27/10/2025